perform the operation on the existing machines (phoenix servers). The second option is to automate the provisioning of new machines running Ubuntu 22.04 and start using those (immutable servers), rather than performing the upgrade on existing machines.

In the next section, I'll talk about automation for building and deploying applications.

1.4.2 Continuous delivery

Continuous delivery is "a software development discipline where you build software in such a way that the software can be released to production at any time."[11] With continuous delivery, teams implement features in short cycles, ensuring that the software can be released at any time reliably. Such a discipline is key to "make high-impact changes frequently and predictably with minimal toil," as per the cloud native definition from the CNCF.

Continuous integration (CI) is a foundational practice in continuous delivery. Developers commit their changes to the mainline (the main branch) continuously (at least once a day). At each commit, the software is automatically compiled, tested, and packaged as executable artifacts (such as JAR files or container images). The idea is to get fast feedback about the software's status after each new change. If an error is detected, it should be immediately fixed to ensure the mainline keeps being a stable foundation for further development.

Continuous delivery (CD) builds on CI and focuses on keeping the mainline always healthy and in a releasable state. After an executable artifact is produced as part of the integration with the mainline, the software is deployed to a production-like environment. It goes through additional tests to assess its *releasability*, such as user acceptance tests, performance tests,